

FixedSense

The Complete Fixed Income Risk Guide

From Zero to Production-Grade Risk Systems

Bonds | Yield Curves | Greeks | VaR | P&L Attribution | Stress Testing

Built for Anant's FixedSense Project

A learning companion — not just theory, but interview-ready depth

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What is a Bond? — The Foundation

Analogy: Think of a bond as lending money to someone with a formal contract. You give them Rs 1000 today. They promise to pay you Rs 50 every year (the coupon), and return your Rs 1000 after 10 years (the maturity). The contract itself — that piece of paper — is the bond. And you can sell that contract to someone else before the 10 years are up.

The 5 Things Every Bond Has

Face Value (Par): The amount the issuer promises to pay back at maturity. Usually Rs 1000 or Rs 100. This is your principal — the original amount lent.

Coupon Rate: The annual interest payment expressed as a percentage of face value. A 5% coupon on Rs 1000 face = Rs 50 per year. This is fixed when the bond is issued and never changes.

Maturity Date: When the issuer returns your face value. Could be 1 year, 5 years, 30 years. Longer maturity = more uncertainty = usually higher yield.

Market Price: What the bond actually trades at TODAY in the market. This changes every second based on interest rates, credit risk, and supply/demand. It can be above face value (premium), below (discount), or equal (par).

Yield: The effective return you earn if you buy at today's market price. This is different from the coupon rate because the price isn't always Rs 1000.

The Golden Rule: Bond prices and yields move in OPPOSITE directions. When interest rates go up, bond prices fall. When rates go down, prices rise. This inverse relationship is the single most important concept in fixed income. Everything else — duration, VaR, risk management — is about quantifying this relationship.

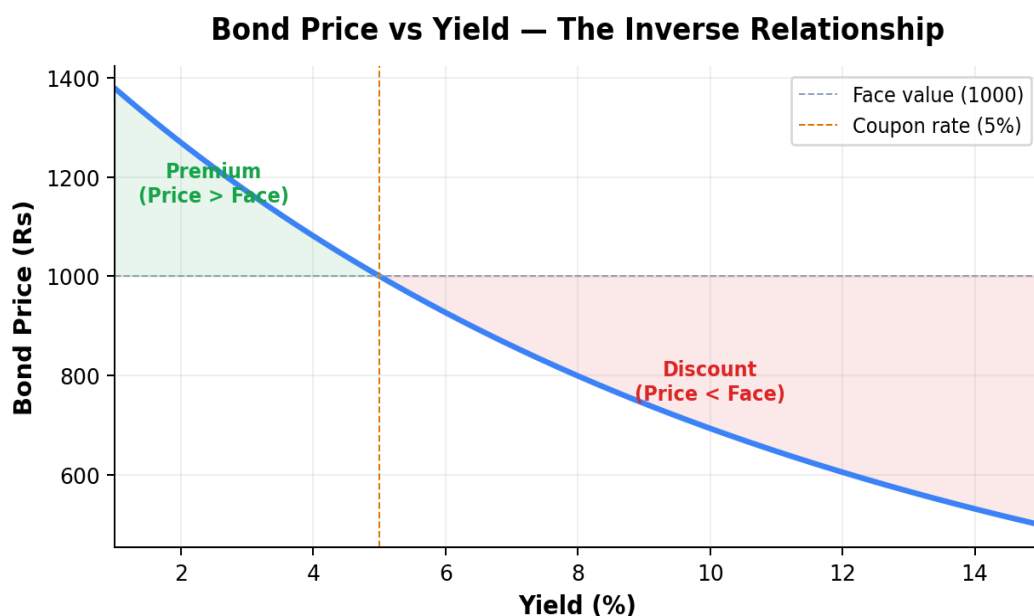


Fig 1.1 — A 5% coupon, 10-year bond. When market yield equals coupon (5%), price = face value. Above 5%, price falls below face (discount). Below 5%, price rises above face (premium).

Why Does This Happen?

Imagine you own a bond paying 5% coupon. Suddenly, new bonds in the market start offering 7%. Nobody wants your 5% bond at full price anymore — why would they, when they can get 7% elsewhere? So your bond's price drops until its effective yield rises to match the market. The opposite happens when new bonds offer only 3% — your 5% bond becomes attractive, and its price rises.

Types of Bonds You'll See in FixedSense

Type	Issuer	Risk Level	Example
Government Bond	Central/State Govt	Lowest (sovereign)	India 10Y G-Sec
Corporate Bond	Companies	Medium to High	Reliance 5Y AAA
Treasury Bill	Government	Lowest, short-term	91-day T-Bill
Zero Coupon Bond	Any	Varies	STRIPS — no coupon, sold at discount

Yield to Maturity (YTM)

Analogy: YTM is like the 'annual return' of a fixed deposit — but for bonds. If you buy a bond today at its market price and hold it until maturity, collecting every coupon and reinvesting them at the same rate, YTM is the single annual return number that makes the math work out.

The coupon rate tells you what the bond was designed to pay. YTM tells you what you'll actually earn given today's price. They're only equal when the bond trades at par (face value).

The YTM Formula

$$\text{Price} = C/(1+y) + C/(1+y)^2 + \dots + C/(1+y)^n + FV/(1+y)^n$$

Where C = coupon payment, y = YTM (what we solve for), n = years to maturity, FV = face value. You can't solve this algebraically — it requires numerical methods (Newton-Raphson or bisection). This is what your code will do.

A Concrete Example

Bond: 5% coupon, 10-year maturity, face value Rs 1000. Market price today: Rs 920.

Since price (920) < face value (1000), we're buying at a discount. This means YTM > coupon rate. Solving the equation gives YTM = 5.87%. The extra 0.87% comes from the capital gain — you paid 920 but get back 1000 at maturity.

If Price is...	Relative to Face	Then YTM is...	Reason
Rs 920	Below (Discount)	> Coupon (5.87%)	Capital gain at maturity adds to return
Rs 1000	Equal (Par)	= Coupon (5.00%)	No gain or loss, just coupon
Rs 1080	Above (Premium)	< Coupon (4.15%)	Capital loss at maturity reduces return

Common Mistake: YTM assumes you reinvest every coupon at the same YTM rate. In reality, rates change — so actual returns will differ. This reinvestment risk is a key limitation interviewers test.

Chapter 3

Duration — Interest Rate Sensitivity

Analogy: Duration is like asking: 'If I push this pendulum, how far will it swing?' For bonds, the push is a change in interest rates, and the swing is the change in price. Higher duration = bigger swing.

There are two types of duration you need to know:

Macaulay Duration

The weighted average time until you receive all your cash flows (coupons + face value). Weights are the present values of each cash flow divided by the bond's total price. A 10-year bond with a 5% coupon has a Macaulay duration around 7.8 years — you effectively get your money back in 7.8 years on average, not 10, because coupons arrive earlier.

Modified Duration

This is the one you actually use for risk. It tells you the percentage price change for a 1% move in yield:

$$\text{Modified Duration} = \text{Macaulay Duration} / (1 + y)$$

$$\text{Percentage Price Change} = -\text{Modified Duration} \times \text{Change in Yield}$$

Example: If modified duration = 7.5 and yields rise by 0.50% (50 basis points):

Price change = $-7.5 \times 0.50\% = -3.75\%$. On a Rs 1000 bond, that's a Rs 37.50 loss.

What Affects Duration: Higher coupon = lower duration (you get money back faster). Longer maturity = higher duration (you wait longer). Higher yield = slightly lower duration. Zero coupon bond has the highest duration — it equals the maturity itself, because there's only one cash flow.

Bond	Coupon	Maturity	Approx Duration	Interpretation
T-Bill	0%	0.25Y	0.25 years	Almost no rate sensitivity
Short-term Corp	6%	2Y	1.9 years	Very low sensitivity
Medium G-Sec	5%	10Y	7.8 years	Moderate — your FixedSense core
Long G-Sec	4%	30Y	18.5 years	Extremely sensitive to rates
Zero coupon 10Y	0%	10Y	10.0 years	Duration = maturity exactly

DV01 — Dollar Value of a Basis Point

Analogy: Duration tells you percentage change. DV01 tells you the ACTUAL MONEY you lose or gain. A portfolio manager doesn't think in percentages — they think 'if rates move 1bp, I lose Rs 4.5 lakhs.' That Rs 4.5 lakhs IS the DV01.

$$\text{DV01} = (\text{Modified Duration} \times \text{Portfolio Value}) / 10,000$$

One basis point (bp) = 0.01% = 0.0001. So DV01 answers: 'For every 0.01% move in rates, how many rupees does my portfolio change by?'

Worked Example

Portfolio value: Rs 100 Crore. Modified Duration: 6.5

$$\text{DV01} = (6.5 \times 100,00,00,000) / 10,000 = \text{Rs } 6,50,000$$

Meaning: every single basis point move in rates costs or earns you Rs 6.5 lakhs. A 25bp rate hike (like an RBI move) = Rs 6.5L x 25 = Rs 1.625 Crore loss.

Why DV01 > Duration in Practice: Traders and risk managers always use DV01 because it's in actual money. 'Duration is 6.5' means nothing to a PM. 'DV01 is Rs 6.5 lakhs' means everything — it's immediately actionable. In your FixedSense dashboard, always show DV01 alongside duration.

Convexity — The Correction Factor

Analogy: Duration is a straight-line approximation of a curved relationship. It's like measuring the slope of a hill at one point and assuming the whole hill has that slope. Convexity corrects for the curvature. For small moves (5-10bps), duration is good enough. For large moves (50-100bps+), you NEED convexity.

$$\text{Better Price Change} = -(\text{Mod Duration} \times dy) + 0.5 \times \text{Convexity} \times dy^2$$

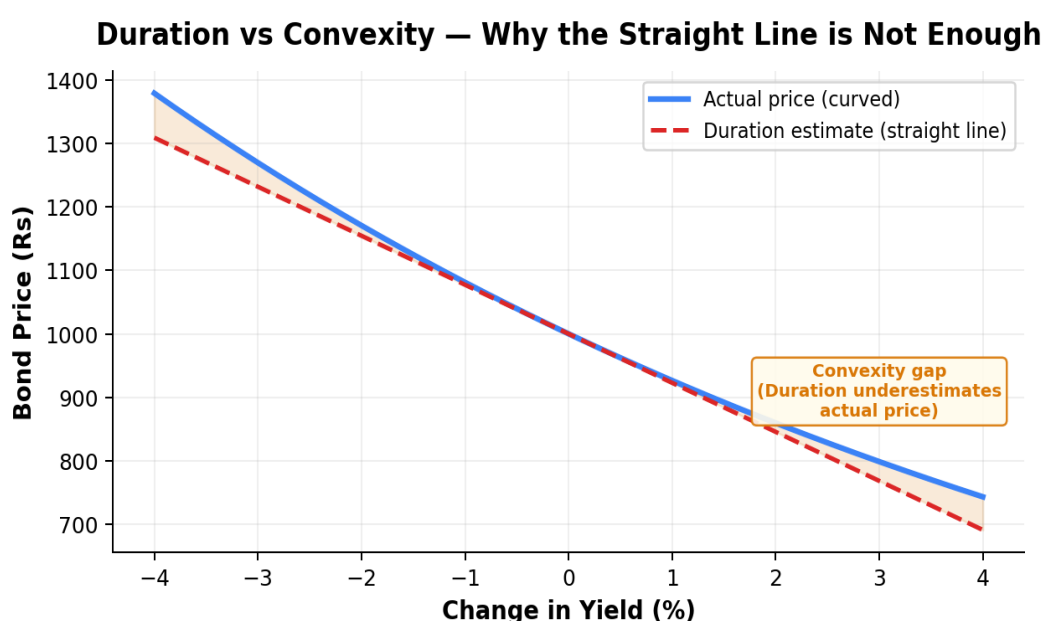


Fig 5.1 — The red dashed line is what duration alone predicts (straight). The blue curve is actual price movement. The orange gap is the convexity correction. Notice: the actual price is ALWAYS above the duration estimate — bonds have positive convexity.

Positive Convexity is Your Friend: Because of convexity, when rates FALL, your bond gains MORE than duration predicts. When rates RISE, your bond loses LESS than duration predicts. This asymmetry is why convexity is considered a good thing — bonds with higher convexity outperform in volatile markets. This is a favorite interview topic.

Example with Numbers

Bond: Modified Duration = 7.5, Convexity = 82. Yield rises by 1% (100bps).

Duration-only estimate: $-7.5 \times 0.01 = -7.50\%$ loss

With convexity: $-7.50\% + 0.5 \times 82 \times (0.01)^2 = -7.50\% + 0.41\% = -7.09\%$ loss

The convexity correction saved you 0.41% — on a Rs 100 Cr portfolio, that's Rs 41 lakhs.

Yield Curves — The Market's Crystal Ball

Analogy: A yield curve is simply a graph: X-axis is 'how long until the bond matures' (tenor), Y-axis is 'what yield does the market demand for that tenor.' It tells you what the market collectively thinks about the future of interest rates. It is arguably the single most important chart in finance.

Three Shapes of the Yield Curve

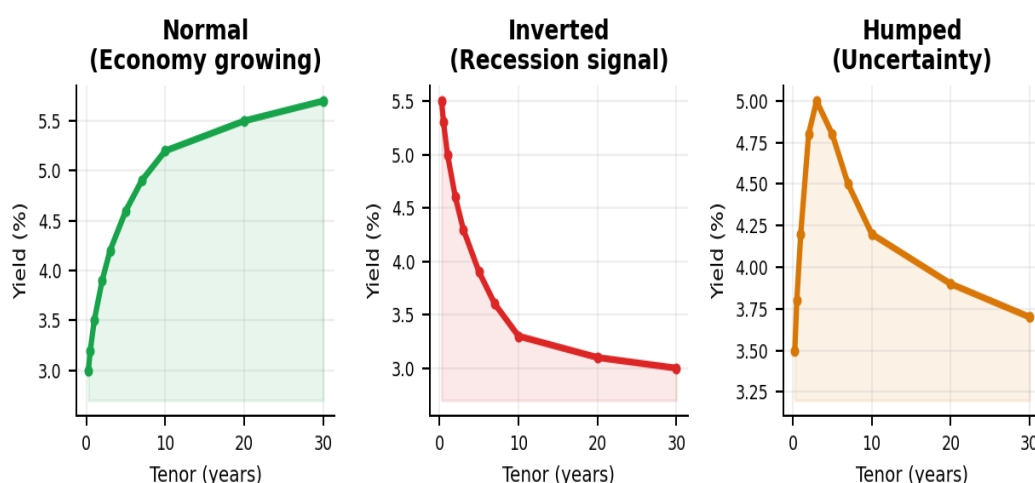


Fig 6.1 — The three yield curve shapes. Normal = longer tenors earn more (compensation for waiting). Inverted = short-term rates higher than long-term (market expects rate cuts, often signals recession). Humped = uncertainty in the medium term.

Spot Rates vs Par Rates

The yield curve you see on Bloomberg is the **par curve** — yields of coupon-paying bonds trading at face value. But for pricing and risk, we need **spot rates** (also called zero rates). A spot rate is the yield on a zero-coupon bond of that maturity. Why? Because spot rates let you discount each cash flow at its own correct rate.

Bootstrapping — Building the Spot Curve

We 'bootstrap' spot rates from the par curve step by step:

Step 1: The 1-year spot rate = 1-year par rate (they're the same for 1 year).

Step 2: For the 2-year par bond, we know the price (100), coupon (say 4%), and the 1-year spot rate. Solve for the 2-year spot rate.

Step 3: Repeat — use all known spot rates to solve for the next one. Each step builds on the previous.

This is exactly what your FixedSense curve engine will do every night when new data arrives.

Forward Rates

Forward rates answer: 'What does the market expect the 1-year rate to be, 2 years from now?' They are implied by spot rates. If the 2-year spot is 4% and the 3-year spot is 4.5%, the implied 1-year forward rate starting in year 2 is roughly 5.5%. The market is pricing in a rate increase. Your FixedSense system can compute these and show whether the market expects rate hikes or cuts.

PCA on Yield Curves — The 3 Factors

Analogy: The yield curve has 9-10 tenor points, each moving independently. That's a lot of variables. PCA (Principal Component Analysis) is a mathematical trick that says: 'Actually, 99% of all yield curve movements can be explained by just 3 patterns.' These 3 patterns are level, slope, and curvature. This is not a simplification — it's what the data actually shows.

This is what separates a production risk system from a toy. Every real quant desk uses PCA factors, not raw tenor points.

PCA Factor Loadings on the Yield Curve

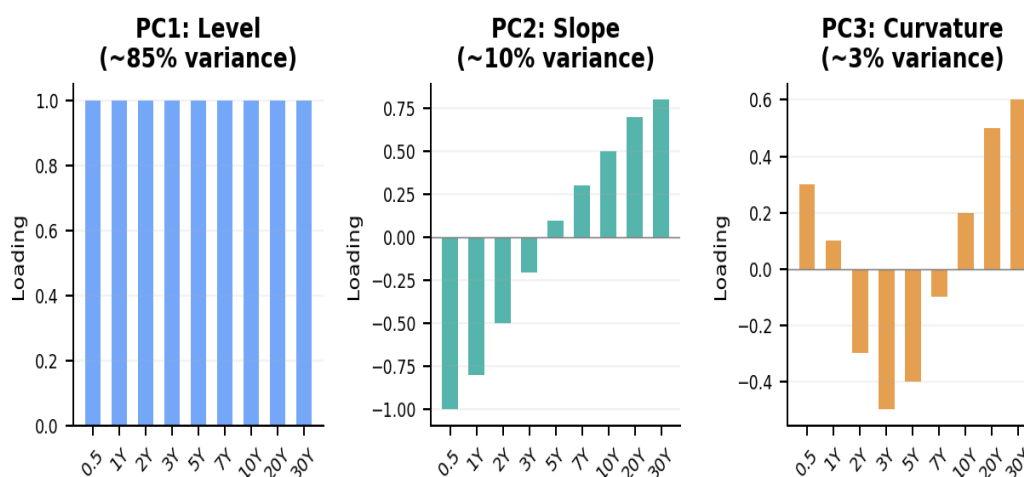


Fig 7.1 — PCA loadings. PC1 (Level): all tenors move equally in the same direction — like the whole curve shifting up/down. PC2 (Slope): short-end moves opposite to long-end — the curve steepens or flattens. PC3 (Curvature): the belly moves opposite to the wings — the curve becomes more or less humped.

How This Works in Practice

Step 1: Collect 2+ years of daily yield curve snapshots (10 tenor points each day).

Step 2: Compute daily changes at each tenor point.

Step 3: Run PCA on the covariance matrix of these changes.

Step 4: The first 3 eigenvectors become your 'factors.' Instead of tracking 10 separate rates, you now track 3 factor exposures.

Why This Matters for FixedSense

When you run Monte Carlo simulation, instead of simulating 10 correlated rate movements (very hard and inaccurate), you simulate 3 independent factor movements (easy and accurate). This is how Goldman and JPMorgan actually do it. Your risk decomposition can then say: '60% of my risk is from level moves, 25% from slope changes, 15% from curvature shifts.'

Interview Gold: If asked 'how would you model yield curve risk?' — answering 'PCA factor model with level, slope, and curvature' immediately signals production-level understanding. Most candidates say 'parallel shift' which is only PC1. You're giving all three.

Key Rate Duration (KR01)

Analogy: DV01 tells you total interest rate risk. KR01 tells you WHERE on the curve that risk sits. It's like DV01, but split into buckets at each tenor point. If your portfolio's KR01 at the 10Y point is Rs 5L but only Rs 0.2L at the 2Y point — you know your risk is concentrated in long-dated bonds.

How KR01 is Computed

At each key tenor point (2Y, 5Y, 10Y, 30Y), bump that specific point by 1 basis point while keeping all other points unchanged. Reprice the entire portfolio. The price change is the KR01 at that tenor.

Sum of all KR01s at each tenor point = Total DV01 of the portfolio (approximately).

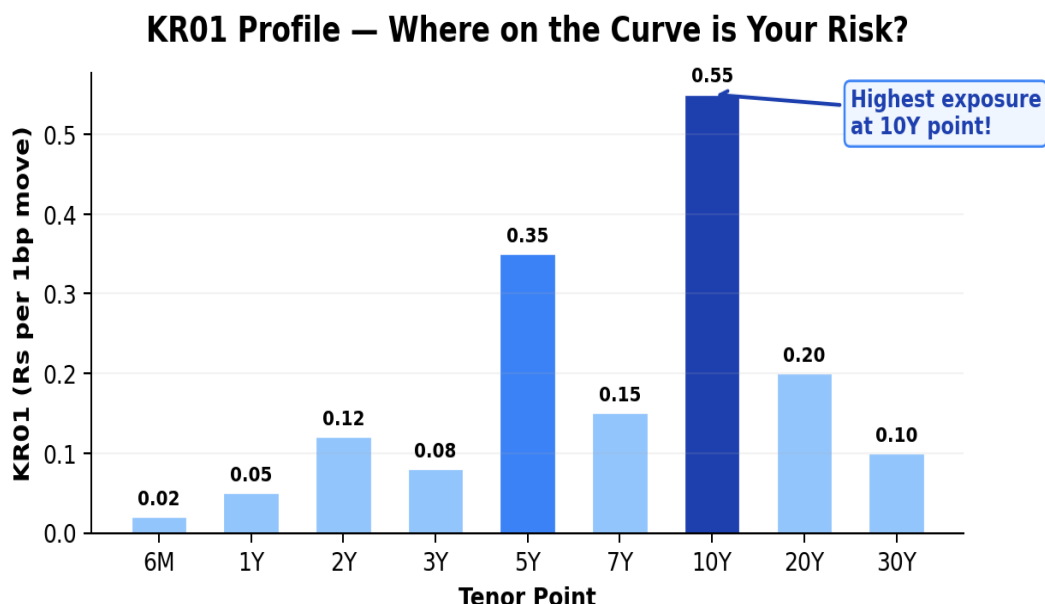


Fig 8.1 — KR01 profile of a sample portfolio. Most risk is at the 10Y point (Rs 0.55 per bp) — meaning rate moves at the 10-year part of the curve matter most. The 2Y exposure is tiny (Rs 0.05) — short-end moves barely affect this portfolio.

Why KR01 Matters More Than DV01

Imagine two portfolios, both with DV01 = Rs 5 lakhs. Portfolio A has all its KR01 at the 2Y point. Portfolio B has all its KR01 at the 30Y point. They look identical by DV01. But Portfolio B is FAR riskier — because 30Y rates are much more volatile and harder to hedge. KR01 reveals this; DV01 hides it.

FixedSense Implementation: Your dashboard should show a KR01 bar chart for the portfolio. When a PM considers adding a bond, the system should show how the KR01 profile changes — does it concentrate risk further at one point, or diversify across the curve?

Chapter 9

Credit Spreads & OAS

Analogy: A government bond is 'risk-free' — the government will almost certainly pay you back. A corporate bond has default risk — the company might go bankrupt. The extra yield the corporate bond pays over the government bond is called the credit spread. It's the market's price for taking on default risk.

$$\text{Credit Spread} = \text{Corporate Bond Yield} - \text{Government Bond Yield (of same maturity)}$$

Example: If a 10Y government bond yields 7% and a 10Y Reliance bond yields 8.2%, the credit spread is 1.2% (120 basis points). This 120bps compensates you for the risk that Reliance might default.

Rating	Typical Spread	Default Risk	Example
AAA	30-80 bps	Negligible	HDFC, Reliance top-tier
AA	60-150 bps	Very Low	Large corporates
A	100-250 bps	Low	Mid-cap corporates
BBB	200-450 bps	Moderate	Smaller companies
Below BBB (Junk)	400-1000+ bps	High	Distressed / speculative

OAS — Option-Adjusted Spread

Some bonds have embedded options (callable bonds can be repaid early by the issuer, puttable bonds let you sell back early). The raw spread includes the value of this option, which distorts the true credit risk. OAS strips out the option value to give you the 'pure' credit spread. In FixedSense, OAS is your true measure of credit risk per bond.

Spread Duration (OAS Duration)

Just like modified duration measures sensitivity to rate changes, spread duration measures sensitivity to spread changes. If OAS duration = 4.5 and spreads widen by 50bps, the bond loses $4.5 \times 0.5\% = 2.25\%$.

Common Mistake: Rate risk and spread risk are DIFFERENT risks that can move independently. In 2008, government yields fell (helping govt bonds) while corporate spreads exploded (crushing corporate bonds). Your FixedSense risk decomposition must separate these two.

Value at Risk (VaR)

Analogy: VaR answers the question every risk manager asks: 'What's the worst-case loss I should NORMALLY expect?' It's like a weather forecast saying 'there's a 95% chance it won't rain more than 50mm today.' VaR says 'there's a 95% chance your portfolio won't lose more than Rs X today.'

Formal definition: VaR at confidence level alpha is the loss threshold that will NOT be exceeded with probability alpha. A 95% 1-day VaR of Rs 10 lakhs means: on 95 out of 100 trading days, your loss will be less than Rs 10 lakhs.

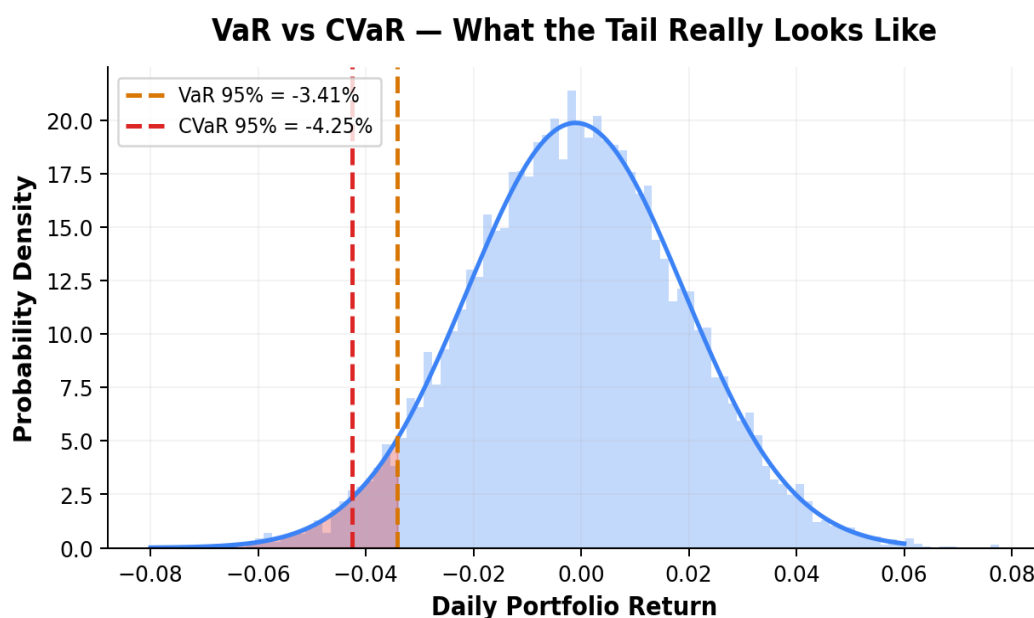


Fig 10.1 — The VaR line is the cutoff. Everything to the left of VaR (the red shaded area) is the worst 5% of days. CVaR (next chapter) averages those worst days.

Three Methods to Compute VaR

Method 1: Parametric (Variance-Covariance)

Assumes returns are normally distributed. $VaR = \text{Portfolio Value} \times z\text{-score} \times \text{daily volatility}$. z-score at 95% = 1.645. Fast to compute but the normality assumption is often wrong — real returns have fat tails.

Method 2: Historical Simulation

Takes the last 500+ days of actual returns, sorts them, and picks the 5th percentile. No distribution assumptions needed — uses real data. But assumes the future will look like the past, which isn't always true.

Method 3: Monte Carlo Simulation

Simulates 10,000+ random scenarios based on a model (could be normal, t-distribution, or factor-based). Sorts the outcomes and picks the 5th percentile. Most flexible — can handle any distribution, any correlation structure, any complexity. This is what FixedSense uses.

Method	Speed	Accuracy	Flexibility	Used At
Parametric	Very fast	Low (assumes normal)	Low	Quick checks
Historical	Fast	Medium	Medium	Regulatory VaR
Monte Carlo	Slow	Highest	Highest	Production risk desks

CVaR / Expected Shortfall

VaR tells you the threshold. CVaR tells you what happens when things go BEYOND that threshold.

Analogy: VaR is like a dam rated to hold water up to 100 meters. CVaR asks: 'When the water overflows, how bad is the flooding on average?' VaR says 'we can handle up to X.' CVaR says 'when X is breached, expect Y on average.' This makes CVaR a much better risk measure for extreme events.

$CVaR(95\%) = \text{Average of all losses in the worst 5\% of scenarios}$

Why CVaR is Better Than VaR

VaR ignores the tail shape. Two portfolios can have the same VaR but completely different tail risk. Portfolio A might lose 5% on its worst days. Portfolio B might lose 40% on its worst days. Same VaR. Very different danger. CVaR captures this difference.

CVaR is 'coherent'. This is a mathematical property. VaR can say that combining two portfolios is riskier than holding them separately (which makes no sense — diversification should reduce risk). CVaR never has this problem. Regulators have noticed.

Basel III/FRTB switched to Expected Shortfall. Regulators now require banks to report Expected Shortfall (same as CVaR) instead of VaR for capital requirements. This is a massive shift. Your FixedSense project computing ES is directly aligned with current regulation.

FixedSense Dashboard Tip: Always show VaR AND CVaR side by side. A portfolio with VaR = Rs 10L and CVaR = Rs 12L has a thin tail (safe). A portfolio with VaR = Rs 10L and CVaR = Rs 35L has a fat tail (dangerous). The ratio CVaR/VaR tells you how 'fat' the tail is.

Monte Carlo Simulation

Analogy: Instead of predicting what WILL happen, simulate 10,000 things that COULD happen and see what the range looks like. It's like playing a video game 10,000 times with slightly different random events each time and counting how often you die.

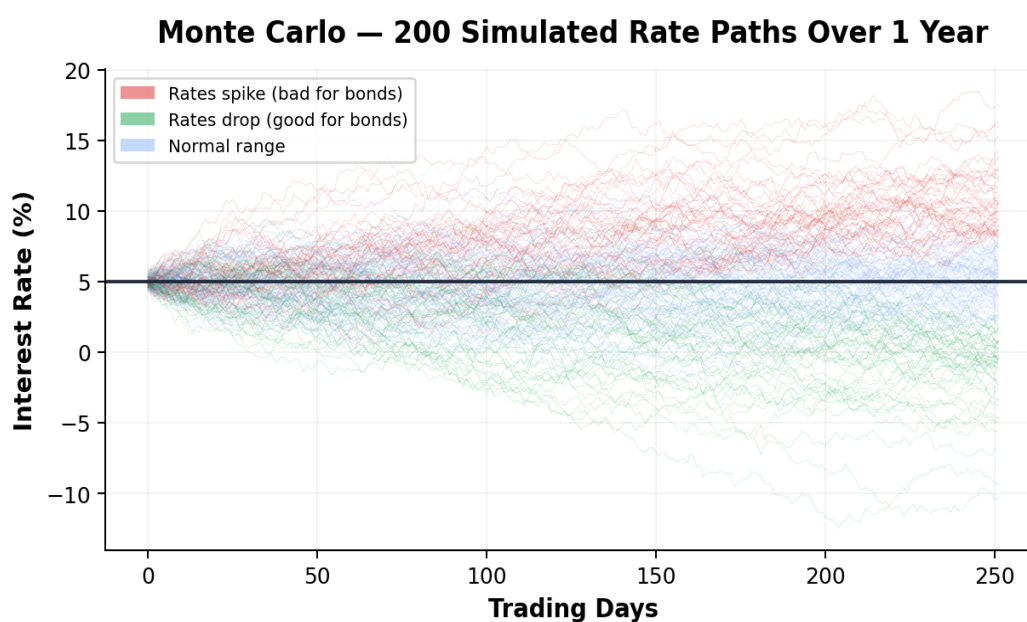


Fig 12.1 — Each thin line is one possible future for interest rates over a year. Most paths (blue) stay near 5%. Some spike above 8% (red — bad for bonds) and some drop below 2% (green — good for bonds). We reprice the portfolio on each path to get a distribution of possible P&L;

How Monte Carlo Works in FixedSense (Step by Step)

Step 1 — Estimate the model. From historical data, estimate: mean daily rate change, volatility of daily changes, and correlations between different tenor points (or use PCA factors).

Step 2 — Generate random scenarios. For each of the 10,000 simulations, generate random factor shocks using the estimated volatilities and correlations. If using PCA, generate 3 independent random numbers (one per factor) and reconstruct the full curve shift.

Step 3 — Apply shocks to the curve. Shift the current yield curve by each simulated shock to get 10,000 different 'tomorrow' yield curves.

Step 4 — Reprice the portfolio. On each of the 10,000 curves, compute the portfolio value using the Greeks (duration, convexity) or full repricing.

Step 5 — Compute risk metrics. Sort the 10,000 P&L outcomes. The 500th worst = VaR (95%). Average of the worst 500 = CVaR (95%).

Common Mistake: Common mistake: using independent random numbers for each tenor point. In reality, rates at different tenors are highly correlated. PCA solves this — simulate 3 independent factors, then reconstruct correlated rate moves. Interviewers WILL test this.

Risk Decomposition — Breaking Down VaR

Analogy: Knowing your portfolio's total VaR is like knowing your total electricity bill. Risk decomposition tells you: AC costs Rs 3000, fridge costs Rs 800, lights cost Rs 400. Now you know WHERE to cut.

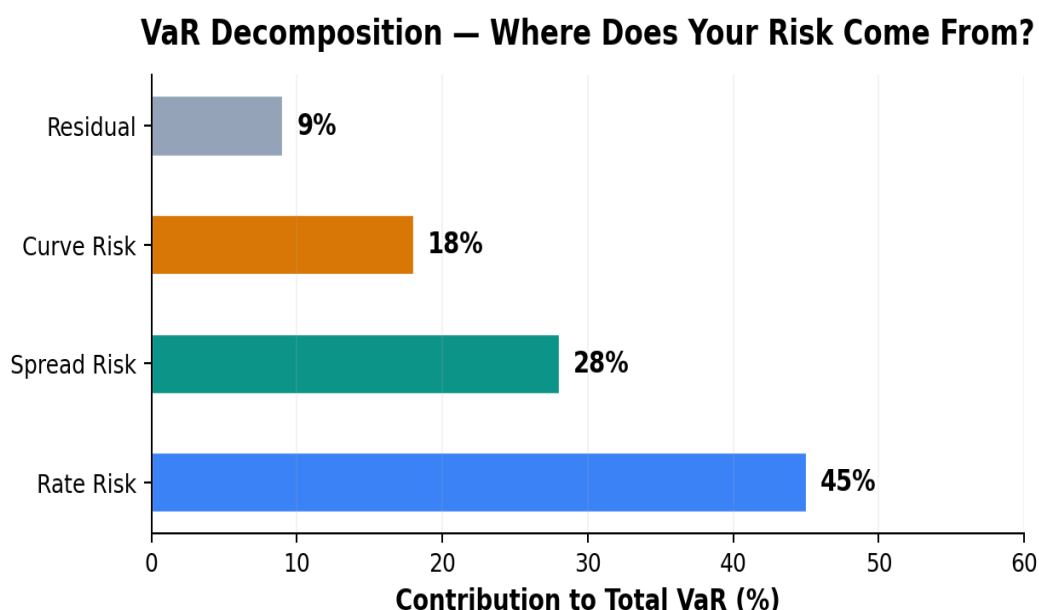


Fig 13.1 — VaR broken into sources. Rate risk dominates (45%) — meaning interest rate moves are the primary driver. Spread risk (28%) is the second concern — credit spread volatility matters. Curve risk (18%) captures non-parallel curve movements.

Three Types of VaR Decomposition

Marginal VaR

How much VaR does each POSITION contribute? If I have 10 bonds, marginal VaR tells me which bond is adding the most risk to the portfolio. This accounts for correlations — a bond that's perfectly hedged by others contributes zero marginal VaR even if it's volatile on its own.

Marginal VaR — Which Bond Adds the Most Risk?

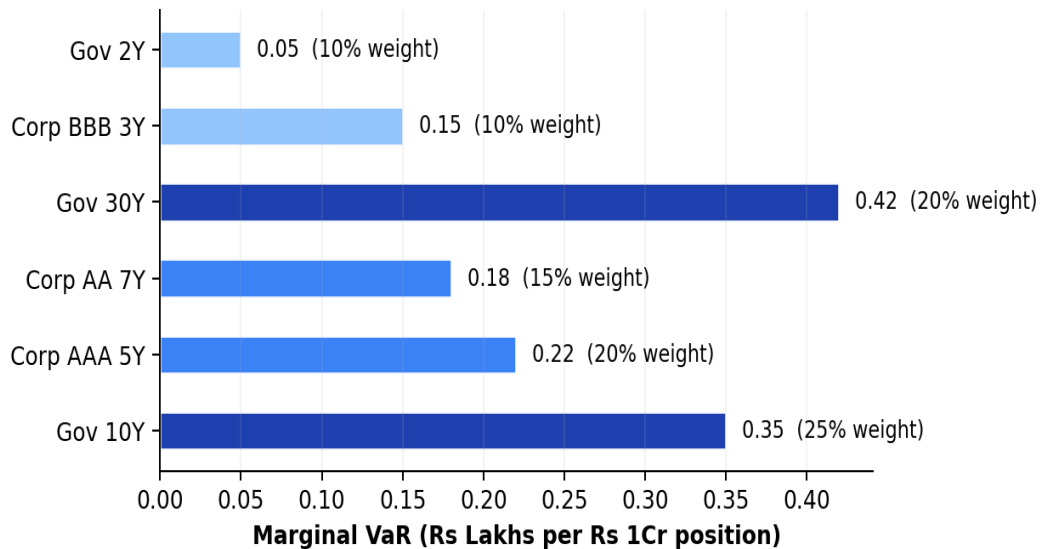


Fig 13.2 — Marginal VaR per bond. The 30Y government bond contributes the most risk (0.42) despite only being 20% of the portfolio — because of its high duration.

Incremental VaR

This answers: 'If I ADD a new bond to the portfolio, how does total VaR change?' This is the trade simulator in FixedSense. A PM says 'I want to buy Rs 10 Cr of Bond X.' The system instantly shows: 'Total VaR increases from Rs 8L to Rs 9.2L, a Rs 1.2L increase.' Or it might decrease — if the new bond hedges existing risk.

Component VaR

Splits total VaR by RISK FACTOR — rate risk, spread risk, curve risk, residual. Sum of all component VaRs = total VaR. This tells the PM: 'Your VaR is Rs 8L, of which Rs 3.6L is rate risk, Rs 2.2L is spread risk, Rs 1.4L is curve risk, and Rs 0.7L is residual.'

P&L; Attribution — The Morning Report

Analogy: Every morning at 8 AM, the portfolio manager looks at one thing: yesterday's P&L; attribution. Not total return. Not VaR. The ATTRIBUTION. 'I made Rs 42L yesterday — was it because of my smart positioning, or did the whole market move in my favor?' P&L; attribution answers this.



Fig 14.1 — P&L; waterfall chart. Starting from previous NAV (10,000), each bar shows what contributed positively or negatively to reach today's NAV. Carry is always positive (coupon income). Rate P&L; is negative (rates rose). Spread P&L; is positive (spreads tightened).

The Four Components of Daily P&L;

Carry P&L;: The coupon income earned by holding the bond for one day. If a bond pays 5% annual coupon on Rs 100 face, daily carry = $5 / 365 = \text{Rs } 0.0137$ per bond. Carry is always positive and predictable. It's the 'safe' part of the return.

Rate P&L;: The gain or loss from changes in the risk-free yield curve. Computed as: $-\text{DV01} \times \text{change in government yield (in bps)}$. If your DV01 is Rs 6.5L and yields rose by 3bps, rate P&L; = $-6.5\text{L} \times 3 = -\text{Rs } 19.5\text{L}$ loss.

Spread P&L;: The gain or loss from changes in credit spreads. Computed using OAS duration. If the spread on your corporate bond tightened by 5bps and spread duration is 4.5, gain = portfolio value $\times 4.5 \times 0.0005$.

Residual: Everything not explained by carry, rates, or spreads. Includes convexity effects, passage of time (theta), model error, and liquidity effects. A large residual signals something the model isn't capturing.

Why This is a Game-Changer for Your CV: P&L; attribution is something 99% of personal projects never implement. Most projects stop at VaR. Building a daily P&L; explain with a waterfall chart shows you understand the PM's actual workflow. This is the feature that will make interviewers lean forward and ask detailed questions.

Stress Testing & Scenario Analysis

VaR handles normal days. Stress testing handles nightmares.

Analogy: VaR is like testing a car at 60 km/h. Stress testing is slamming it into a wall at 120 km/h. You need both. Banks failed in 2008 not because their VaR was wrong on normal days — but because they never stress tested for what happened on abnormal days.

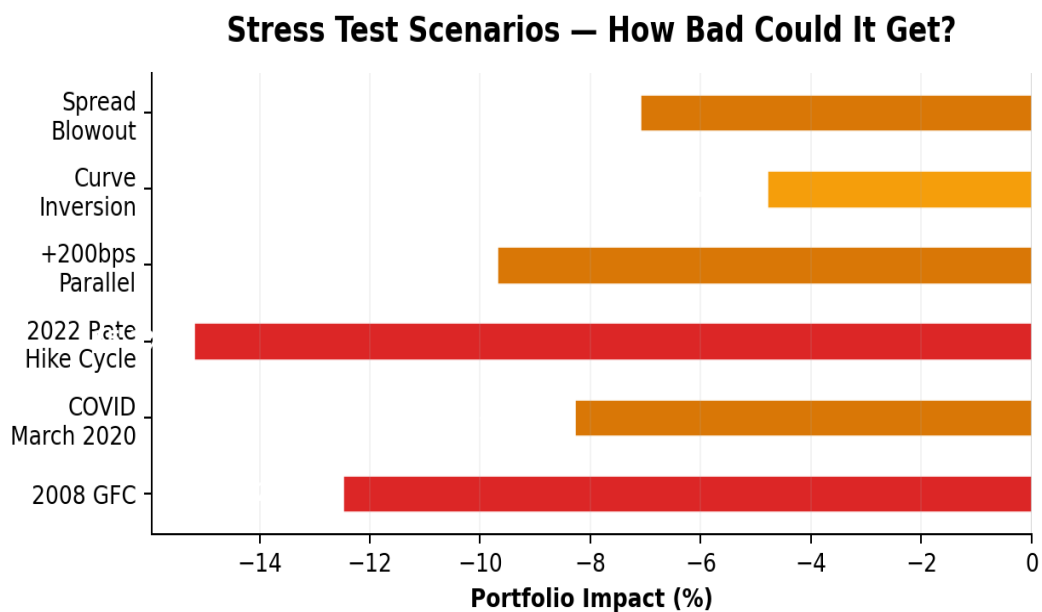


Fig 15.1 — Six stress scenarios applied to a sample portfolio. Historical replays (2008, COVID, 2022) use actual market movements from those periods. Hypothetical scenarios are manually constructed shocks.

Three Types of Stress Tests in FixedSense

Historical Replay

Take the actual daily market changes from a crisis period and apply them to your current portfolio. 'If 2008 happened again with my current holdings, I'd lose 12.5%.' This is powerful because it uses real data, not assumptions.

Hypothetical Scenarios

Manually constructed shocks: 'What if rates rise 200bps in one week?' 'What if the yield curve inverts by 100bps?' These don't need historical precedent — they test specific risks the PM is worried about.

Correlated Multi-Asset Shocks

The most realistic type. In a real crisis, everything moves together. Rates spike, spreads blow out, FX crashes, and liquidity evaporates simultaneously. Your scenario engine should shock rates, spreads, and FX in a correlated way using historical crisis correlations.

Implementation Detail: For each crisis period (e.g., September-November 2008), extract daily changes in: government yields at each tenor point, credit spreads by rating bucket, and any FX rates. Store these as 'scenario vectors.' To replay, apply the entire sequence of daily shocks to the current portfolio and track the cumulative P&L; path.

Regulatory Framework — Basel & FRTB

Banks don't compute VaR just because it's useful. They're legally required to. Understanding the regulatory framework shows you understand WHY risk systems exist, not just HOW they work.

Basel III — The Current Standard

After the 2008 crisis, regulators realized banks were under-capitalized. Basel III requires banks to hold a minimum amount of capital based on their risk. The riskier the trading book, the more capital must be held. This capital requirement is computed using risk models — including VaR.

FRTB — The New Framework (Fundamental Review of the Trading Book)

FRTB replaces the old Basel VaR-based capital rules. Key changes:

Expected Shortfall replaces VaR: Banks must now use ES (97.5% confidence) instead of VaR. This is exactly what FixedSense computes.

Stressed ES: In addition to normal ES, banks must compute ES using data from a stressed period (like 2008). This is the sVaR module in FixedSense.

Liquidity Horizons: Different risk factors have different time horizons. FX risk can be hedged in 10 days, credit spread risk takes 40-120 days. FRTB requires scaling ES to these horizons.

Risk Factor Eligibility Test (RFET): If you can't prove a risk factor has real price data, you can't use your internal model for it — you must use a simpler, more conservative standardised approach.

Metric	Basel II/III	FRTB (New)	FixedSense
Core metric	VaR (99%, 10-day)	ES (97.5%)	Both computed
Stress component	Stressed VaR	Stressed ES	Historical replay
Risk factors	Aggregated	Granular buckets	PCA factors + KR01
Backtesting	250-day window	Enhanced P&L test	Daily P&L attribution

Interview Questions & Answers

These questions are organized by topic and difficulty. For each, think about the answer before reading it.

Bond Basics & Yield

Q: What happens to bond prices when interest rates rise?

They fall. Bond prices and yields have an inverse relationship. When new bonds offer higher yields, existing bonds with lower coupons become less attractive, so their price drops until their effective yield matches the market.

Q: A bond has a 6% coupon and trades at Rs 1050. Is its YTM above or below 6%?

Below 6%. Since the price is ABOVE face value (premium), you're paying extra — so your effective return is less than the coupon rate. You'll incur a capital loss of Rs 50 at maturity, reducing your total return below the 6% coupon.

Q: What is reinvestment risk and why does it matter for YTM?

YTM assumes every coupon is reinvested at the same YTM rate. In reality, rates change — so actual reinvestment happens at different rates. Higher coupon bonds have more reinvestment risk because more of their return comes from reinvested coupons rather than the final payment.

Q: Explain the difference between clean price and dirty price.

Clean price excludes accrued interest. Dirty price = clean price + accrued interest. Bonds trade on clean price (to avoid price jumps on coupon dates) but settle on dirty price (the actual amount paid). Accrued interest = coupon x (days since last coupon / days in coupon period).

Duration & DV01

Q: Rank these by duration: 30Y zero coupon, 10Y 5% coupon, 2Y 8% coupon.

30Y zero coupon (highest — 30 years, all cash flow at maturity) > 10Y 5% coupon (moderate — around 7.8 years) > 2Y 8% coupon (lowest — around 1.9 years). Higher maturity and lower coupon both increase duration.

Q: Your portfolio DV01 is Rs 5 lakhs. RBI hikes rates by 25bps. What's your approximate loss?

Loss = DV01 x basis points = 5,00,000 x 25 = Rs 1,25,00,000 = Rs 1.25 Crore. This is an approximation using duration only — convexity would slightly reduce the actual loss.

Q: Why is modified duration not sufficient for large rate moves?

Modified duration assumes a linear price-yield relationship, but the actual relationship is curved (convex). For small moves (<10bps), the linear approximation is fine. For larger moves (50-100bps+), the convexity term becomes significant. Duration always overestimates losses and underestimates gains — convexity corrects this.

Q: What's the difference between DV01 and duration? When would you use each?

Duration gives percentage price change; DV01 gives dollar/rupee price change per basis point. Duration is scale-independent (useful for comparing bonds). DV01 is scale-dependent (useful for actual P&L; and hedging). Traders use DV01 because they think in money. Analysts use duration because they compare across different portfolio sizes.

Yield Curves & PCA

Q: What does an inverted yield curve tell you?

Short-term rates are higher than long-term rates. This typically signals the market expects rate cuts ahead — often because a recession is coming. Historically, yield curve inversions have preceded most US recessions by 12-18 months. However, it's a signal, not a guarantee.

Q: Explain how you would bootstrap a spot curve from par yields.

Start with the shortest maturity — the 1Y par rate equals the 1Y spot rate. For the 2Y: take a 2Y par bond (price=100, coupon=par rate). Discount the first coupon using the known 1Y spot. Solve for the 2Y spot rate that makes the present value equal 100. Repeat for each successive tenor, always using previously computed spot rates for earlier coupons.

Q: Why use PCA for yield curve modeling instead of simulating each tenor independently?

Tenor points are highly correlated — they don't move independently. Simulating them independently ignores these correlations and produces unrealistic scenarios. PCA extracts the 3 dominant patterns (level, slope, curvature) that explain ~99% of variance. You simulate 3 independent factors instead of 10 correlated tenors — mathematically correct and computationally efficient.

Q: What do the three PCA factors represent and how much variance does each explain?

PC1 (Level, ~85%): parallel shift — all rates move up or down together. PC2 (Slope, ~10%): short rates move opposite to long rates — the curve steepens or flattens. PC3 (Curvature, ~3%): the belly of the curve moves opposite to the wings — humping or flattening of the mid-section. Together they explain ~98-99% of all yield curve movements.

KR01 & Credit Spreads

Q: Why is KR01 more useful than DV01 for risk management?

DV01 assumes a parallel shift — all tenors move by the same amount. This rarely happens. KR01 shows risk at each specific tenor point, revealing concentration. Two portfolios with identical DV01 can have very different KR01 profiles — one might be concentrated at 30Y (very risky) while the other is spread across tenors (diversified).

Q: How would you hedge a portfolio that has high KR01 at the 10Y point?

Short 10Y government bond futures, or enter a 10Y interest rate swap (pay fixed, receive floating). This offsets the 10Y exposure specifically without affecting other tenor exposures. If you used a 2Y hedge instead, you'd only reduce 2Y KR01 and leave the 10Y risk untouched.

Q: What causes credit spreads to widen?

Multiple factors: deteriorating issuer credit quality, rising default expectations, market-wide risk aversion (flight to quality), reduced liquidity in corporate bond markets, economic downturns, or sector-specific shocks. In March 2020 (COVID), investment-grade spreads tripled in two weeks purely from panic and liquidity withdrawal.

Q: What is OAS and why can't we just use the raw spread?

Raw spread = bond yield - government yield. For callable bonds, part of this spread compensates for the call option — not credit risk. OAS strips out the option value by running a model (binomial tree) to isolate pure credit compensation. For non-callable bonds, OAS approximately equals the raw spread.

VaR, CVaR & Monte Carlo

Q: What are the limitations of VaR?

Four key limitations: (1) It says nothing about tail shape — losses beyond VaR could be 1% or 50%. (2) It's not subadditive — portfolio VaR can exceed sum of individual VaRs, penalizing diversification. (3) Parametric VaR assumes normality, which understates tail risk. (4) All methods assume some stationarity — the future might differ from the past/model.

Q: Why did FRTB switch from VaR to Expected Shortfall?

ES (CVaR) addresses VaR's two biggest flaws: it captures tail shape (averages all losses beyond the threshold) and it's coherent/subadditive (diversification always reduces or maintains risk). FRTB uses ES at 97.5% confidence, which captures approximately the same quantile as 99% VaR but provides more information about the tail.

Q: In your Monte Carlo simulation, how do you handle correlations between risk factors?

Two approaches: (1) Cholesky decomposition of the covariance matrix — generate independent random numbers, then multiply by the Cholesky matrix to introduce correlations. (2) PCA-based — simulate independent factor scores, then reconstruct correlated rate changes using the factor loadings. PCA is preferred because it's dimensionally reduced and more stable.

Q: How many simulations do you need and how do you know it's enough?

Typically 10,000+ for stable VaR and 50,000+ for stable CVaR (tail metrics need more samples). Convergence test: run the simulation multiple times with different random seeds. If VaR changes by more than 2-3% between runs, you need more paths. For FixedSense, 10,000 is a good balance of accuracy and speed.

Q: How would you backtest your VaR model?

Compare predicted VaR against actual realized P&L; over 250+ trading days. Count the number of 'exceptions' (days where actual loss exceeded predicted VaR). For 95% VaR, expect ~12-13 exceptions in 250 days. Too few means the model is too conservative. Too many means it underestimates risk. Basel uses a traffic light system: 0-4 exceptions = green, 5-9 = yellow, 10+ = red.

Risk Decomposition & P&L; Attribution

Q: Explain marginal vs incremental vs component VaR.

Marginal VaR: how much VaR does each existing position currently contribute? (Sums to total VaR). Component VaR: same concept but split by risk factor (rate, spread, curve) instead of by position. Incremental VaR: how would total VaR change if I ADD or REMOVE a position? (Forward-looking, used for trade decisions). Marginal and component explain the present; incremental helps decide the future.

Q: Walk me through how you'd compute daily P&L; attribution.

Step 1: Record yesterday's portfolio (positions, prices, curve). Step 2: Compute today's curve. Step 3: Carry = sum of daily coupon accrual. Step 4: Rate P&L; = $-DV01 \times (\text{change in government yield in bps})$. Step 5: Spread P&L; = $-\text{spread_DV01} \times (\text{change in OAS in bps})$. Step 6: Residual = actual total P&L; - carry - rate P&L; - spread P&L.; The residual captures convexity, theta, and model error.

Q: Your residual P&L; is consistently large. What could cause this?

Several possibilities: (1) Convexity effects not captured by the linear duration approximation — need to add the convexity term. (2) Theta/time decay not modeled — bonds 'pull to par' as they approach maturity. (3) Incorrect curve or spread data — check the data feed. (4) Missing risk factors — perhaps FX or liquidity is moving the portfolio. (5) Stale prices — if bonds are illiquid, prices may not update daily.

Stress Testing & Regulatory

Q: How do you design a hypothetical stress scenario?

Start with a narrative (e.g., 'Fed hikes 300bps in 6 months'). Translate to market shocks: +300bps across the curve, but with steepening (front-end more than back-end). Add correlated effects: credit spreads widen 100-200bps for IG, 400+ for HY. FX: USD strengthens 10%. Apply all shocks simultaneously and measure portfolio impact. Document the narrative and assumptions — regulators require this.

Q: What is the difference between historical and hypothetical stress testing?

Historical: replay actual market movements from a past crisis on today's portfolio. Advantage: realistic, data-driven, no assumptions needed. Limitation: can only test what has already happened. Hypothetical: construct custom scenarios. Advantage: can test unprecedented events. Limitation: requires assumptions about correlations and magnitudes that may be wrong.

Q: What is FRTB and why should your risk system care?

FRTB (Fundamental Review of the Trading Book) is the new regulatory standard replacing Basel 2.5 for market risk capital. Key changes: VaR replaced by ES, liquidity horizons added, stricter model approval via P&L; attribution tests, and desk-level capital assignment. Any production risk system must compute ES, stressed ES, and pass P&L; attribution backtesting — exactly what FixedSense does.

Q: Explain the P&L; attribution test in FRTB.

FRTB requires that your risk model's predicted daily P&L; closely matches actual realized P&L;. The test compares the theoretical P&L; (from your risk factors) against the hypothetical P&L; (from full repricing). If they diverge too often, the desk loses permission to use internal models and must use the standardised approach — which requires much more capital. This is why P&L; attribution quality matters.

FixedSense Architecture & Design

Q: Why did you choose S3 + Parquet over a traditional database?

For time-series financial data, S3+Parquet offers: columnar storage (read only needed columns — perfect for analytics), schema evolution (add fields without migrations), infinite scale at low cost, native integration with Python/Pandas/Spark, and partition pruning by date. A traditional RDBMS would struggle with the append-heavy, read-heavy pattern of daily risk computation. Parquet on S3 is the industry standard at quant shops.

Q: Why use PCA-based Monte Carlo instead of simpler parametric VaR?

Parametric VaR assumes normality, which understates tail risk. It also only captures parallel shifts. PCA-based Monte Carlo uses empirical factor distributions (can model fat tails), captures non-parallel curve moves (slope, curvature), handles nonlinear payoffs via full repricing, and provides the full return distribution — not just a single quantile. The computational cost is worth the accuracy gain.

Q: How would you extend FixedSense to handle a portfolio of 500+ bonds?

Vectorize all pricing and Greek computations using NumPy. Pre-compute and cache the spot curve and factor loadings (they're the same for all bonds). Use approximate repricing (duration + convexity) instead of full repricing for Monte Carlo — with periodic full-reprice validation. Partition S3 data by date and instrument type. For intraday streaming, only recompute Greeks for bonds whose prices actually moved.

You've Reached the End

This guide covers every financial concept behind FixedSense.

Next step: Build it. Start with the yield curve engine (Chapter 7) — everything else depends on it.

FixedSense Learning Guide v1.0 — Built for Anant